Experiment 4

Statistical Hypothesis Testing Using Python

Subject: Data Science Lab | Dataset: Titanic | Mapped CO(s): CO2

1. Aim

To perform statistical hypothesis testing on the Titanic dataset using Python and interpret the results using p-values.

2. Key Concepts & Definitions

Hypothesis Testing

A method to decide whether sample data provides enough evidence to support a claim about a population.

Null Hypothesis (H0)

Assumes no significant difference or relationship. Example: mean ages are equal.

Alternative Hypothesis (H1)

Assumes there is a significant difference or relationship. Example: mean ages differ.

Significance Level (alpha)

Threshold for deciding significance. Commonly alpha = 0.05 (5%).

$$\alpha = 0.05$$

p-value

Probability of observing a result at least as extreme as the sample result, assuming H0 is true.

Decision Rule

If p < alpha, reject H0. If p >= alpha, fail to reject H0.

Independent Sample t-test

Compares means of two independent groups (e.g. survived vs not survived).

Chi-Square Test of Independence

Tests association between two categorical variables (e.g. Sex vs Survived).

3. Hypotheses for This Experiment (Age)

H0: Mean age of survived passengers = mean age of non-survived passengers

$$H_0 : \mu_{\text{survived}} = \mu_{\text{not survived}}$$

H1: Mean ages are not equal

$$H_1 : \mu_{\text{survived}} \neq \mu_{\text{not survived}}$$

4. Important Formulae

(A) Two-sample t-statistic (unequal variances)

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

(B) Decision rule

$$\text{Reject } H_0 \text{ if } p < \alpha$$

$$\text{Fail to reject } H_0 \text{ if } p \geq \alpha$$

(C) Chi-square statistic

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

O = observed frequency, E = expected frequency under independence.

5. Algorithm

- 1. Import Pandas, NumPy, SciPy, Matplotlib, Seaborn
- 2. Load Titanic dataset
- 3. Select Age and Survived; drop missing ages
- 4. Split into survived and not-survived groups
- 5. Perform independent t-test (scipy.stats.ttest_ind)
- 6. Compare p-value with alpha = 0.05 and conclude

6. Symbol Glossary

Symbol	Meaning
H0	Null hypothesis
H1	Alternative hypothesis
alpha	Significance level
p	p-value
t	t-statistic
x-bar	Sample mean
s	Sample standard deviation
n	Sample size
chi^2	Chi-square statistic

7. Viva Q&A

Q: What is H0?

A: Null hypothesis - assumes no significant difference.

Q: What is p-value?

A: Probability of extreme results under H0.

Q: When do we reject H0?

A: When p-value < alpha (usually 0.05).

Q: What is an independent t-test?

A: Compares means of two independent groups.

Q: What is a Chi-square test?

A: Tests association between two categorical variables.

8. Conclusion

Hypothesis testing was applied on Titanic Age (and Fare) using an independent t-test, and association between Sex and Survived was tested using Chi-square. Decisions were made by comparing p-values with the chosen significance level.